Reference

BlueXP backup and recovery

NetApp
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Reference

BlueXP backup and recovery policy configuration settings

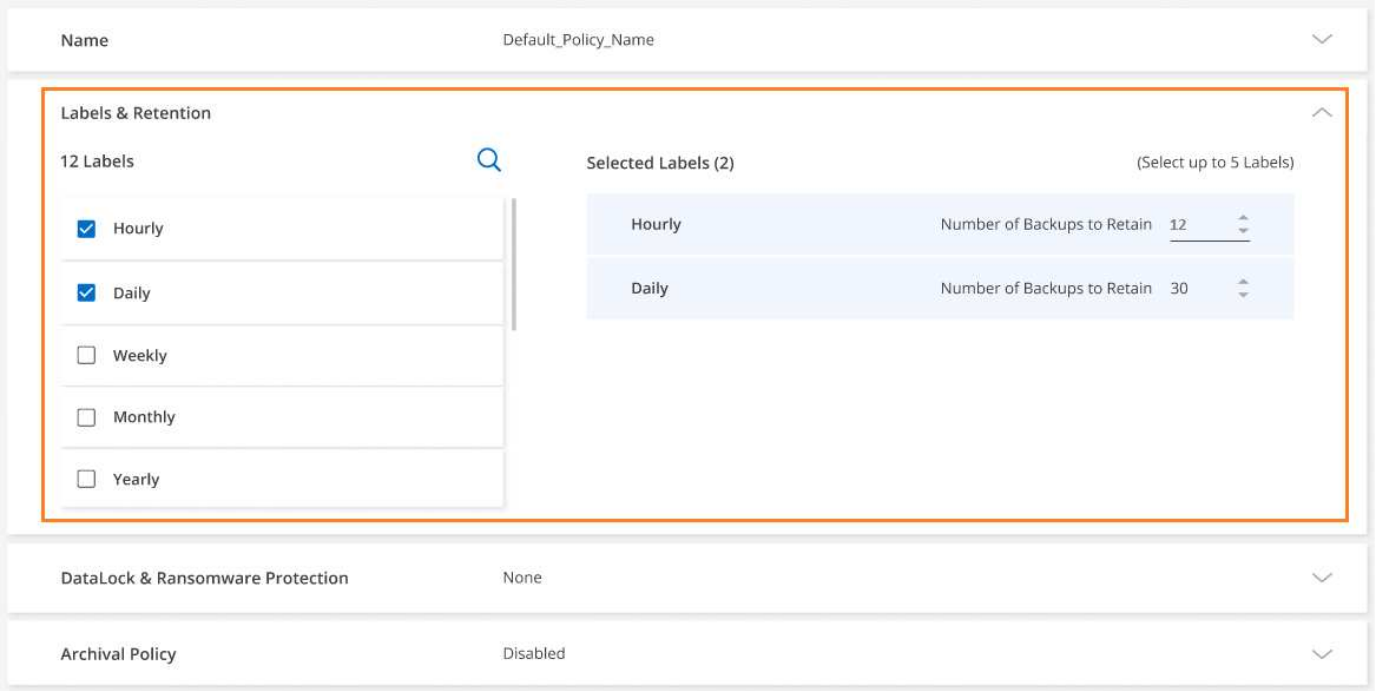
BlueXP backup and recovery enables you to create backup policies with a variety of configuration settings for your on-prem ONTAP and Cloud Volumes ONTAP systems.

> These policy settings are relevant for backup to object storage only. None of these settings affect your Snapshot or replication policies. Similar policy settings for Snapshots and replications will be added in the future.

Backup schedules

BlueXP backup and recovery enables you to create multiple backup policies with unique schedules for each working environment (cluster). You can assign different backup policies to volumes that have different recovery point objectives (RPO).

Each backup policy provides a section for *Labels & Retention* that you can apply to your backup files. Note that the Snapshot policy applied to the volume must be one of the policies recognized by BlueXP backup and recovery or backup files will not be created.



There are two parts of the schedule; the Label and the Retention value:

- The **label** defines how often a backup file is created (or updated) from the volume. You can select among the following types of labels:
  - You can choose one, or a combination of, **hourly**, **daily**, **weekly**, **monthly**, and **yearly** timeframes.
  - You can select one of the system-defined policies that provide backup and retention for 3 months, 1 year, or 7 years.
  - If you have created custom backup protection policies on the cluster using ONTAP System Manager or

the ONTAP CLI, you can select one of those policies.

- The **retention** value defines how many backup files for each label (timeframe) are retained. Once the maximum number of backups have been reached in a category, or interval, older backups are removed so you always have the most current backups. This also saves you storage costs because obsolete backups don't continue to take up space in the cloud.

For example, say you create a backup policy that creates 7 **weekly** and 12 **monthly** backups:

- each week and each month a backup file is created for the volume
- at the 8th week, the first weekly backup is removed, and the new weekly backup for the 8th week is added (keeping a maximum of 7 weekly backups)
- at the 13th month, the first monthly backup is removed, and the new monthly backup for the 13th month is added (keeping a maximum of 12 monthly backups)

Note that Yearly backups will be deleted automatically from the source system after being transferred to object storage. This default behavior can be changed in the Advanced Settings page for the Working Environment.

DataLock and Ransomware protection

BlueXP backup and recovery provides support for DataLock and Ransomware protection for your volume backups. These features enable you to lock your backup files and scan them to detect possible ransomware on the backup files. This is an optional setting that you can define in your backup policies when you want extra protection for your volume backups for a cluster.

Both of these features protect your backup files so that you'll always have a valid backup file to recover data from in case of a ransomware attack attempt on your backups. It's also helpful to meet certain regulatory requirements where backups need to be locked and retained for a certain period of time. When the DataLock and Ransomware Protection option is enabled, the cloud bucket that is provisioned as a part of BlueXP backup and recovery activation will have object locking and object versioning enabled.

See the DataLock and Ransomware protection blog for more details.

This feature does not provide protection for your source volumes; only for the backups of those source volumes. Use NetApp Cloud Insights and Cloud Secure, or some of the anti-ransomware protections provided from ONTAP to protect your source volumes.

- If you plan to use DataLock and Ransomware protection, you must enable it when creating your first backup policy and activating BlueXP backup and recovery for that cluster.
- DataLock and Ransomware protection can't be disabled for a cluster once it has been configured. Do not enable this feature on a cluster to try it out.
- When BlueXP scans a backup file for ransomware when restoring volume data, you'll incur extra egress costs from your cloud provider to access the contents of the backup file.

What is DataLock

DataLock protects your backup files from being modified or deleted for a certain period of time - also called *immutable storage*. This functionality uses technology from the object storage provider for "object locking." The period of time that the backup file is locked (and retained) is called the DataLock Retention Period. It is based on the backup policy schedule and retention setting that you defined; plus a 14-day buffer. Any DataLock retention policy that is less than 30 days is rounded up to 30 days minimum.

Be aware that old backups are deleted after the DataLock Retention Period expires, not after the backup policy

retention period expires.

Let’s look at some examples of how this works:

- If you create a Monthly backup schedule with 12 retentions, each backup is locked for 12 months (plus 14 days) before it is deleted.
- If you create a backup policy that creates 30 daily, 7 weekly, 12 monthly backups there will be three locked retention periods. The "30 daily" backups would be retained for 44 days (30 days plus 14 days buffer), the "7 weekly" backups would be retained for 9 weeks (7 weeks plus 14 days), and the "12 monthly" backups would be retained for 12 months (plus 14 days).
- If you create an Hourly backup schedule with 24 retentions, you might think that backups are locked for 24 hours. However, since that is less than the minimum of 30 days, each backup will be locked and retained for 44 days (30 days plus 14 days buffer).

  You can see in this last case that if each backup file is locked for 44 days, you’ll end up with many more backup files than would typically be retained with an hourly/24 retentions policy. Usually, when BlueXP backup and recovery creates the 25th backup file it would delete the oldest backup to keep the maximum retentions at 24 (based on the policy). The DataLock retention setting overrides the policy retention setting from your backup policy in this case. This could affect your storage costs as your backup files will be saved in the object store for a longer period of time.

What is Ransomware protection

Ransomware protection scans your backup files to look for evidence of a ransomware attack. The detection of ransomware attacks is performed using a checksum comparison. If potential ransomware is identified in a new backup file versus the previous backup file, that newer backup file is replaced by the most recent backup file that does not show any signs of a ransomware attack. (The file that was identified as having a ransomware attack is deleted 1 day after it has been replaced.)

Ransomware scans happen at 3 points in the backup and restore process:

- When a backup file is created

  The scan is not performed on the backup file when it is first written to cloud storage, but when the **next** backup file is written. For example, if you have a weekly backup schedule set for Tuesday, on Tuesday the 14th a backup is created. Then on Tuesday the 21st another backup is created. The ransomware scan is run on the backup file from the 14th at this time.

- When you attempt to restore data from a backup file

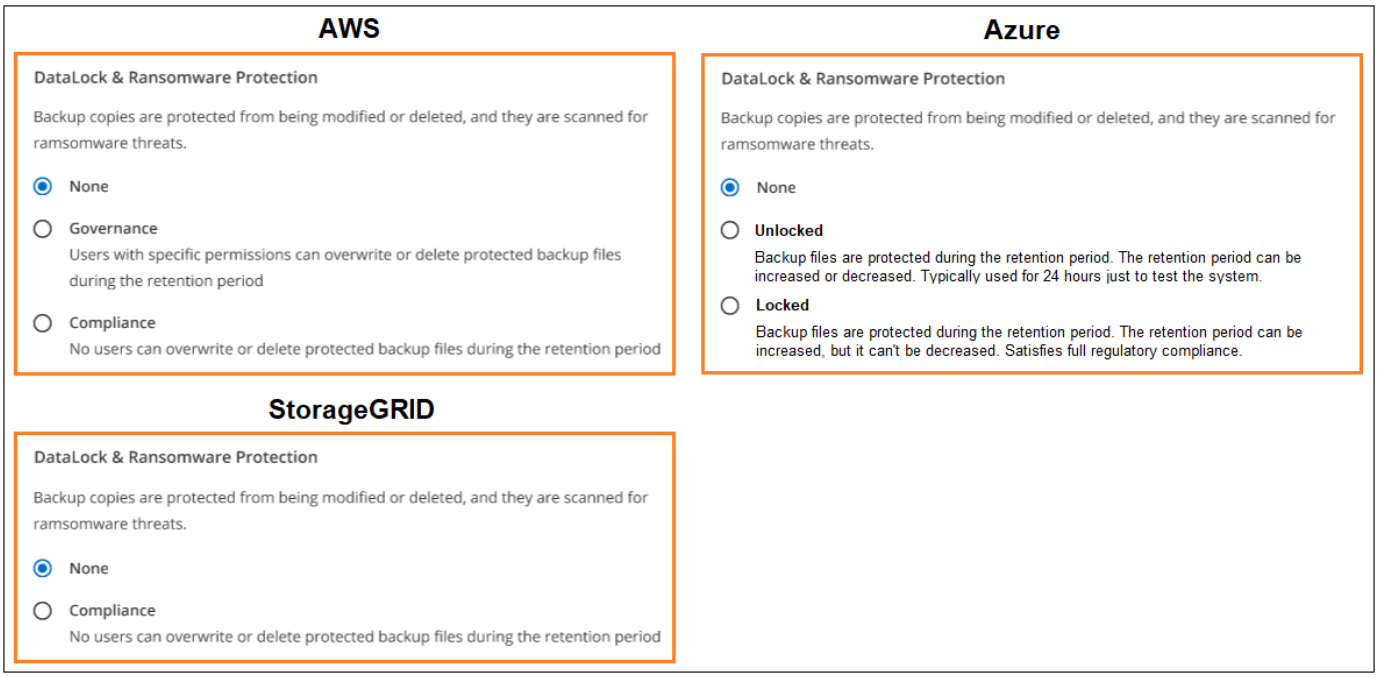
  You can choose to run a scan before restoring data from a backup file, or skip this scan.

- Manually

  You can run an on-demand ransomware protection scan at any time to verify the health of a specific backup file. This can be useful if you’ve had a ransomware issue on a particular volume and you want to verify that the backups for that volume are not affected.

DataLock and Ransomware Protection settings

Each backup policy provides a section for *DataLock and Ransomware Protection* that you can apply to your backup files.



You can choose from the following settings for each backup policy:

AWS

- **None** (Default)

  DataLock protection and ransomware protection are disabled.

- **Governance**

  DataLock is set to *Governance* mode where users with `s3:BypassGovernanceRetention` permission (see below) can overwrite or delete backup files during the retention period. Ransomware protection is enabled.

- **Compliance**

  DataLock is set to *Compliance* mode where no users can overwrite or delete backup files during the retention period. Ransomware protection is enabled.

StorageGRID

- **None** (Default)

  DataLock protection and ransomware protection are disabled.

- **Compliance**

  DataLock is set to *Compliance* mode where no users can overwrite or delete backup files during the retention period. Ransomware protection is enabled.

Supported working environments and object storage providers

You can enable DataLock and Ransomware protection on ONTAP volumes from the following working

environments when using object storage in the following public and private cloud providers. Additional cloud providers will be added in future releases.

Source Working Environment	Backup File Destination
Cloud Volumes ONTAP in AWS	Amazon S3
On-premises ONTAP system	Amazon S3 NetApp StorageGRID

Requirements

- For AWS:
  - Your clusters must running ONTAP 9.11.1 or greater
  - The Connector can be deployed in the cloud or on your premises
  - The following S3 permissions must be part of the IAM role that provides the Connector with permissions. They reside in the "backupS3Policy" section for the resource "arn:aws:s3:::netapp-backup-*":
    - s3:GetObjectVersionTagging
    - s3:GetBucketObjectLockConfiguration
    - s3:GetObjectVersionAcl
    - s3:PutObjectTagging
    - s3:DeleteObject
    - s3:DeleteObjectTagging
    - s3:GetObjectRetention
    - s3:DeleteObjectVersionTagging
    - s3:PutObject
    - s3:GetObject
    - s3:PutBucketObjectLockConfiguration
    - s3:GetLifecycleConfiguration
    - s3:GetBucketTagging
    - s3:DeleteObjectVersion
    - s3:ListBucketVersions
    - s3:ListBucket
    - s3:PutBucketTagging
    - s3:GetObjectTagging
    - s3:PutBucketVersioning
    - s3:PutObjectVersionTagging
    - s3:GetBucketVersioning
    - s3:GetBucketAcl
    - s3:BypassGovernanceRetention
    - s3:PutObjectRetention

        - s3:GetBucketLocation
        - s3:GetObjectVersion

        View the full JSON format for the policy where you can copy and paste required permissions.

- For StorageGRID:
    - Your clusters must running ONTAP 9.11.1 or greater
    - Your StorageGRID systems must be running 11.6.0.3 or greater
    - The Connector must be deployed on your premises (it can be installed in a site with or without internet access)
    - The following S3 permissions must be part of the IAM role that provides the Connector with permissions:
        - s3:GetObjectVersionTagging
        - s3:GetBucketObjectLockConfiguration
        - s3:GetObjectVersionAcl
        - s3:PutObjectTagging
        - s3:DeleteObject
        - s3:DeleteObjectTagging
        - s3:GetObjectRetention
        - s3:DeleteObjectVersionTagging
        - s3:PutObject
        - s3:GetObject
        - s3:PutBucketObjectLockConfiguration
        - s3:GetLifecycleConfiguration
        - s3:GetBucketTagging
        - s3:DeleteObjectVersion
        - s3:ListBucketVersions
        - s3:ListBucket
        - s3:PutBucketTagging
        - s3:GetObjectTagging
        - s3:PutBucketVersioning
        - s3:PutObjectVersionTagging
        - s3:GetBucketVersioning
        - s3:GetBucketAcl
        - s3:PutObjectRetention
        - s3:GetBucketLocation
        - s3:GetObjectVersion

Restrictions

- The DataLock and Ransomware protection feature is not available if you have configured archival storage in the backup policy.
- The DataLock option you select when activating BlueXP backup and recovery must be used for all backup policies for that cluster.
- You cannot use multiple DataLock modes on a single cluster.
- If you enable DataLock, all volume backups will be locked. You can't mix locked and non-locked volume backups for a single cluster.
- DataLock and Ransomware protection is applicable for new volume backups using a backup policy with DataLock and Ransomware protection enabled. You can't enable this feature after BlueXP backup and recovery has been activated.
- FlexGroup volumes can use DataLock and Ransomware protection only when using ONTAP 9.13.1 or greater.

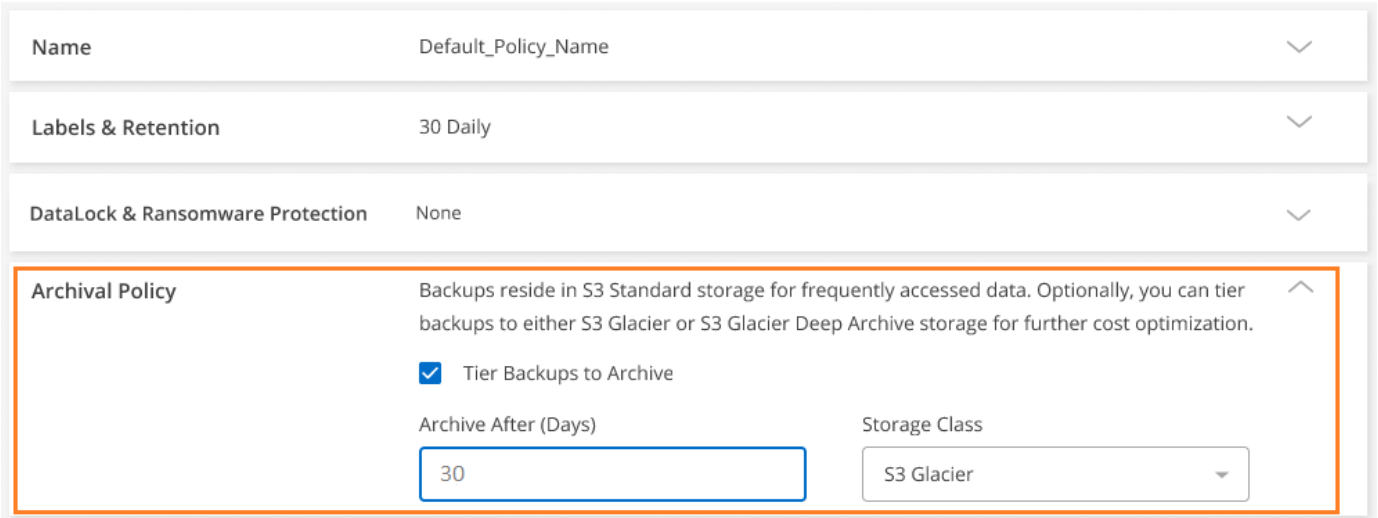
Archival storage settings

When using AWS, Azure, or Google cloud storage, you can move older backup files to a less expensive archival storage class or access tier after a certain number of days. You can also choose to send your backup files to archival storage immediately without being written to standard cloud storage. Just enter **0** as the "Archive After Days" to send your backup file directly to archival storage. This can be especially helpful for users who rarely need to access data from cloud backups or users who are replacing a backup to tape solution.

Data in archival tiers can't be accessed immediately when needed, and will require a higher retrieval cost, so you'll need to consider how often you may need to restore data from backup files before deciding to archive your backup files.

- Even if you select "0" to send all data blocks to archival cloud storage, metadata blocks are always written to standard cloud storage.
- Archival storage can't be used if you have enabled DataLock.
- You can't change the archival policy after selecting **0** days (archive immediately).

Each backup policy provides a section for *Archival Policy* that you can apply to your backup files.



- In AWS, backups start in the *Standard* storage class and transition to the *Standard-Infrequent Access* storage class after 30 days.

  If your cluster is using ONTAP 9.10.1 or greater, you can tier older backups to either *S3 Glacier* or *S3 Glacier Deep Archive* storage. Learn more about AWS archival storage.

  - If you select no archive tier in your first backup policy when activating BlueXP backup and recovery, then *S3 Glacier* will be your only archive option for future policies.
  - If you select *S3 Glacier* in your first backup policy, then you can change to the *S3 Glacier Deep Archive* tier for future backup policies for that cluster.
  - If you select *S3 Glacier Deep Archive* in your first backup policy, then that tier will be the only archive tier available for future backup policies for that cluster.
- In StorageGRID, backups are associated with the *Standard* storage class.

  If your on-prem cluster is using ONTAP 9.12.1 or greater, and your StorageGRID system is using 11.4 or greater, you can archive older backup files to public cloud archival storage.

  - For AWS, you can tier backups to AWS *S3 Glacier* or *S3 Glacier Deep Archive* storage. Learn more about AWS archival storage.

    Learn more about archiving backup files from StorageGRID.

AWS S3 archival storage classes and restore retrieval times

BlueXP backup and recovery supports two S3 archival storage classes and most regions.

Supported S3 archival storage classes for BlueXP backup and recovery

When backup files are initially created they're stored in S3 *Standard* storage. This tier is optimized for storing data that's infrequently accessed; but that also allows you to access it immediately. After 30 days the backups transition to the S3 *Standard-Infrequent Access* storage class to save on costs.

If your source clusters are running ONTAP 9.10.1 or greater, you can choose to tier backups to either *S3 Glacier* or *S3 Glacier Deep Archive* storage after a certain number of days (typically more than 30 days) for further cost optimization. Data in these tiers can't be accessed immediately when needed, and will require a higher retrieval cost, so you need to consider how often you may need to restore data from these archived backup files. See the section about restoring data from archival storage.

- If you select no archive tier in your first backup policy when activating BlueXP backup and recovery, then *S3 Glacier* will be your only archive option for future policies.
- If you select *S3 Glacier* in your first backup policy, then you can change to the *S3 Glacier Deep Archive* tier for future backup policies for that cluster.
- If you select *S3 Glacier Deep Archive* in your first backup policy, then that tier will be the only archive tier available for future backup policies for that cluster.

Note that when you configure BlueXP backup and recovery with this type of lifecycle rule, you must not configure any lifecycle rules when setting up the bucket in your AWS account.

Learn about S3 storage classes.

Restoring data from archival storage

While storing older backup files in archival storage is much less expensive than Standard or Standard-IA storage, accessing data from a backup file in archive storage for restore operations will take a longer amount of time and will cost more money.

How much does it cost to restore data from Amazon S3 Glacier and Amazon S3 Glacier Deep Archive?

There are 3 restore priorities you can choose when retrieving data from Amazon S3 Glacier, and 2 restore priorities when retrieving data from Amazon S3 Glacier Deep Archive. S3 Glacier Deep Archive costs less than S3 Glacier:

Archive Tier	Restore Priority & Cost		
	High	**Standard**	**Low**
S3 Glacier	Fastest retrieval, highest cost	Slower retrieval, lower cost	Slowest retrieval, lowest cost
S3 Glacier Deep Archive		Faster retrieval, higher cost	Slower retrieval, lowest cost

Each method has a different per-GB retrieval fee and per-request fee. For detailed S3 Glacier pricing by AWS Region, visit the Amazon S3 pricing page.

How long will it take to restore my objects archived in Amazon S3 Glacier?

There are 2 parts that make up the total restore time:

- **Retrieval time**: The time to retrieve the backup file from archive and place it in Standard storage. This is sometimes called the "rehydration" time. The retrieval time is different depending on the restore priority you choose.

Archive Tier	Restore Priority & Retrieval Time		
	High	**Standard**	**Low**
S3 Glacier	3-5 minutes	3-5 hours	5-12 hours
S3 Glacier Deep Archive		12 hours	48 hours

- **Restore time**: The time to restore the data from the backup file in Standard storage. This time is no different than the typical restore operation directly from Standard storage - when not using an archival tier.

For more information about Amazon S3 Glacier and S3 Glacier Deep Archive retrieval options, refer to the Amazon FAQ about these storage classes.

Azure archival tiers and restore retrieval times

BlueXP backup and recovery supports one Azure archival access tier and most regions.

Supported Azure Blob access tiers for BlueXP backup and recovery

When backup files are initially created they’re stored in the *Cool* access tier. This tier is optimized for storing data that’s infrequently accessed; but when needed, can be accessed immediately.

If your source clusters are running ONTAP 9.10.1 or greater, you can choose to tier backups from *Cool* to *Azure Archive* storage after a certain number of days (typically more than 30 days) for further cost optimization. Data in this tier can't be accessed immediately when needed, and will require a higher retrieval cost, so you need to consider how often you may need to restore data from these archived backup files. See the next section about restoring data from archival storage.

Note that when you configure BlueXP backup and recovery with this type of lifecycle rule, you must not configure any lifecycle rules when setting up the container in your Azure account.

Learn about Azure Blob access tiers.

Restoring data from archival storage

While storing older backup files in archival storage is much less expensive than Cool storage, accessing data from a backup file in Azure Archive for restore operations will take a longer amount of time and will cost more money.

How much does it cost to restore data from Azure Archive?

There are two restore priorities you can choose when retrieving data from Azure Archive:

- **High**: Fastest retrieval, higher cost
- **Standard**: Slower retrieval, lower cost

Each method has a different per-GB retrieval fee and per-request fee. For detailed Azure Archive pricing by Azure Region, visit the Azure pricing page.

> The High priority is not supported when restoring data from Azure to StorageGRID systems.

How long will it take to restore my data archived in Azure Archive?

There are 2 parts that make up the restore time:

- **Retrieval time**: The time to retrieve the archived backup file from Azure Archive and place it in Cool storage. This is sometimes called the "rehydration" time. The retrieval time is different depending on the restore priority you choose:
  - **High**: < 1 hour
  - **Standard**: < 15 hours
- **Restore time**: The time to restore the data from the backup file in Cool storage. This time is no different than the typical restore operation directly from Cool storage - when not using an archival tier.

For more information about Azure Archive retrieval options, refer to this Azure FAQ.

Google archival storage classes and restore retrieval times

BlueXP backup and recovery supports one Google archival storage class and most regions.

Supported Google archival storage classes for BlueXP backup and recovery

When backup files are initially created they're stored in *Standard* storage. This tier is optimized for storing data that's infrequently accessed; but that also allows you to access it immediately.

If your on-prem cluster is using ONTAP 9.12.1 or greater, you can choose to tier older backups to *Archive* storage in the BlueXP backup and recovery UI after a certain number of days (typically more than 30 days) for further cost optimization. Data in this tier will require a higher retrieval cost, so you need to consider how often you may need to restore data from these archived backup files. See the section about restoring data from archival storage.

Note that when you configure BlueXP backup and recovery with this type of lifecycle rule, you must not configure any lifecycle rules when setting up the bucket in your Google account.

Learn about Google storage classes.

Restoring data from archival storage

While storing older backup files in Archive storage is much less expensive than Standard storage, accessing data from a backup file in Archive storage for restore operations will take a slightly longer amount of time and will cost more money.

How much does it cost to restore data from Google Archive?

For detailed Google Cloud Storage pricing by region, visit the Google Cloud Storage pricing page.

How long will it take to restore my objects archived in Google Archive?

There are 2 parts that make up the total restore time:

- **Retrieval time**: The time to retrieve the backup file from Archive and place it in Standard storage. This is sometimes called the "rehydration" time. Unlike the "coldest" storage solutions provided by other cloud providers, your data is accessible within milliseconds.
- **Restore time**: The time to restore the data from the backup file in Standard storage. This time is no different than the typical restore operation directly from Standard storage - when not using an archival tier.

Restore BlueXP backup and recovery data in a dark site

When using BlueXP backup and recovery in a site with no internet access, known as "private mode", the BlueXP backup and recovery configuration data is backed up to the StorageGRID or ONTAP S3 bucket where your backups are being stored. If you have an issue with the BlueXP Connector host system in the future, you can deploy a new Connector and restore the critical BlueXP backup and recovery data.

Note that when you use BlueXP backup and recovery in a SaaS environment where the BlueXP Connector is deployed at your cloud provider, or on your own host system that has internet access, all the important BlueXP backup and recovery configuration data is backed up and protected in the cloud. If you have an issue with the Connector, just create a new Connector and add your working environments and the backup details are automatically restored.

There are 2 types of data that are backed up:

- BlueXP backup and recovery database - contains a listing of all the volumes, backup files, backup policies, and configuration information.
- Indexed Catalog files - contains detailed indexes that are used for Search & Restore functionality that make your searches very quick and efficient when looking for volume data that you want to restore.

This data is backed up once per day at midnight, and a maximum of 7 copies of each file are retained. If the

Connector is managing multiple on-premises ONTAP working environments, the BlueXP backup and recovery files will be located in the bucket of the working environment that was activated first.

> No volume data is ever included in the BlueXP backup and recovery database or Indexed Catalog files.

Restore BlueXP backup and recovery data to a new Connector

If your on-premises Connector has a catastrophic failure, you'll need to install a new Connector, and then restore the BlueXP backup and recovery data to the new Connector.

There are 4 tasks you'll need to perform to return your BlueXP backup and recovery system to a working state:

- Install a new BlueXP Connector
- Restore the BlueXP backup and recovery database
- Restore the Indexed Catalog files
- Rediscover all of your on-prem ONTAP systems and StorageGRID systems to the BlueXP UI

Once you verify that your system is back in a working order, we recommend that you create new backup files.

What you'll need

You'll need to access the most recent database and index backups from the StorageGRID or ONTAP S3 bucket where your backup files are being stored:

- BlueXP backup and recovery MySQL database file

  This file is located in the following location in the bucket `netapp-backup-<GUID>/mysql_backup/`, and it is named `CBS_DB_Backup_<day>_<month>_<year>.sql`.

- Indexed Catalog backup zip file

  This file is located in the following location in the bucket `netapp-backup-<GUID>/catalog_backup/`, and it is named `Indexed_Catalog_DB_Backup_<db_name>_<day>_<month>_<year>.zip`.

Install a new Connector on a new on-premises Linux host

When installing a new BlueXP Connector, make sure you download the same release of software as you had installed on the original Connector. Periodic changes to the BlueXP backup and recovery database structure may make newer software releases incompatible with the original database backups. You can upgrade the Connector software to the most current version after restoring the Backup database.

1. Install the BlueXP Connector on a new on-premises Linux host
2. Log into BlueXP using the admin user credentials that you just created.

Restore the BlueXP backup and recovery database

1. Copy the MySQL backup from the backup location to the new Connector host. We'll use the example file name "CBS_DB_Backup_23_05_2023.sql" below.
2. Copy the backup into the MySQL docker container using the following command:

```
docker cp CBS_DB_Backup_23_05_2023.sql ds_mysql_1:/.
```

3. Enter the MySQL container shell using the following command:

```
docker exec -it ds_mysql_1 sh
```

4. In the container shell, deploy the "env".
5. You'll need the MySQL DB password, so copy the value of the key "MYSQL_ROOT_PASSWORD".
6. Restore the BlueXP backup and recovery MySQL DB using the following command:

```
mysql -u root -p cloud_backup < CBS_DB_Backup_23_05_2023.sql
```

7. Verify that the BlueXP backup and recovery MySQL DB has been restored correctly using the following SQL commands:

```
mysql -u root -p cloud_backup
```

Enter the password.

```
mysql> show tables;
mysql> select * from volume;
```

Check if the volumes that are shown are the same as those that existed in your original environment.

Restore the Indexed Catalog files

1. Copy the Indexed Catalog backup zip file (we'll use the example file name "Indexed_Catalog_DB_Backup_catalogdb1_23_05_2023.zip") from the backup location to the new Connector host in the "/opt/application/netapp/cbs" folder.
2. Unzip the "Indexed_Catalog_DB_Backup_catalogdb1_23_05_2023.zip" file using the following command:

```
unzip Indexed_Catalog_DB_Backup_catalogdb1_23_05_2023.zip -d catalogdb1
```

3. Run the **ls** command to make sure that the folder "catalogdb1" has been created with the subfolders "changes" and "snapshots" underneath.

Discover your ONTAP clusters and StorageGRID systems

1. Discover all the on-prem ONTAP working environments that were available in your previous environment. This includes the ONTAP system you have used as an S3 server.
2. Discover your StorageGRID systems.

Set up the StorageGRID environment details

Add the details of the StorageGRID system associated with your ONTAP working environments as they were set up on the original Connector setup using the BlueXP APIs.

You'll need to perform these steps for each ONTAP system that is backing up data to StorageGRID.

1. Extract the authorization token using the following oauth/token API.

```
curl 'http://10.193.192.202/oauth/token' -X POST -H 'User-Agent:
Mozilla/5.0 (Macintosh; Intel Mac OS X 10.15; rv:100101 Firefox/108.0'
-H 'Accept: application/json' -H 'Accept-Language: en-US,en;q=0.5' -H
'Accept-Encoding: gzip, deflate' -H 'Content-Type: application/json' -d
'{"username":admin@netapp.com,"password":"Netapp@123","grant_type":"pass
word"}
> '
```

   This API will return a response like the following. You can retrieve the authorization token as shown below.


```
{"expires_in":21600,"access_token":"eyJhbGciOiJSUzI1NiIsInR5cCI6IkpXVCIs
ImtpZCI6IjJlMGFiZjRiIn0eyJzdWIiOiJvY2NtYXV0aHwxIiwiYXVkIjpbImh0dHBzOi8vY
XBpLmNsb3VkLm5ldGFwcC5jb20iXSwiaHR0cDovL2Nsb3VkLm5ldGFwcC5jb20vZnVsbF9uY
W1lIjoiYWRtaW4iLCJodHRwOi8vY2xvdWQubmV0YXBwLmNvbS9lbWFpbCI6ImFkbWluQG5ld
GFwcC5jb20iLCJzY29wZSI6Im9wZW5pZCBwcm9maWxlIiwiaWF0IjoxNjcyNzM2MDIzLCJle
HAiOjE2NzI3NTc2MjMsImlzcyI6Imh0dHA6Ly9vY2NtYXV0aDo4NDIwLyJ9CJtRpRDY23Pok
yLg1if67bmgnMcYxdCvBOY-ZUYWzhrWbbY_hqUH4T-
114v_pNDsPyNDyWqHaKizThdjjHYHxm56vTz_Vdn4NqjaBDPwN9KAnC6Z88WA1cJ4WRQqj5y
kODNDmrv5At_f9HHp0-xVMyHqywZ4nNFalMvAh4xESc5jfoKOZc-
IOQdWm4F4LHpMzs4qFzCYthTuSKLYtqSTUrZB81-o-ipvrOqSo1iwIeHXZJJV-
UsWun9daNgiYd_wX-4WWJViGEnDzzwOKfUoUoe1Fg3ch--7JFkFl-
rrXDOjk1sUMumN3WHV9usp1PgBE5HAcJPrEBm0ValSZcUbiA"}
```


2. Extract the Working Environment ID and the X-Agent-Id using the tenancy/external/resource API.

```
curl -X GET
http://10.193.192.202/tenancy/external/resource?account=account-
DARKSITE1 -H 'accept: application/json' -H 'authorization: Bearer
eyJhbGciOiJSUzI1NiIsInR5cCI6IkpXVCIsImtpZCI6IjJlMGFiZjRiIn0eyJzdWIiOiJvY
2NtYXV0aHwxIiwiYXVkIjpbImh0dHBzOi8vYXBpLmNsb3VkLm5ldGFwcC5jb20iXSwiaHR0c
DovL2Nsb3VkLm5ldGFwcC5jb20vZnVsbF9uYW1lIjoiYWRtaW4iLCJodHRwOi8vY2xvdWQub
mV0YXBwLmNvbS9lbWFpbCI6ImFkbWluQG5ldGFwcC5jb20iLCJzY29wZSI6Im9wZW5pZCBwc
m9maWxlIiwiaWF0IjoxNjcyNzIyNzEzLCJleHAiOjE2NzI3NDQzMTMsImlzcyI6Imh0dHA6L
y9vY2NtYXV0aDo4NDIwLyJ9X_cQF8xttD0-S7sU2uph2cdu_kN-
fLWpdJJX98HODwPpVUitLcxV28_sQhuopjWobozPelNISf7KvMqcoXc5kLDyX-
yE0fH9gr4XgkdswjWcNvw2rRkFzjHpWrETgfqAMkZcAukV4DHuxogHWh6-
DggB1NgPZT8A_szHinud5W0HJ9c4AaT0zC-
sp81GaqMahPf0KcFVyjbBL4krOewgKHGFo_7ma_4mF39B1LCj7Vc2XvUd0wCaJvDMjwp19-
KbZqmmBX9vDnYp7SSxC1hHJRDStcFgJLdJHtowweNH2829KsjEGBTTcBdO8SvIDtctNH_GAx
wSgMT3zUfwaOimPw'
```

This API will return a response like the following. The value under the "resourceIdentifier" denotes the *WorkingEnvironment Id* and the value under "agentId" denotes *x-agent-id*.

3. Update the BlueXP backup and recovery database with the details of the StorageGRID system associated with the Working Environments. Make sure to enter the Fully Qualified Domain Name of the StorageGRID, as well as the Access-Key and Storage-Key as shown below:

```
curl -X POST 'http://10.193.192.202/account/account-
DARKSITE1/providers/cloudmanager_cbs/api/v1/sg/credentials/working-
environment/OnPremWorkingEnvironment-pMtZND0M' \
> --header 'authorization: Bearer
eyJhbGciOiJSUzI1NiIsInR5cCI6IkpXVCIsImtpZCI6IjJlMGFiZjRiIn0eyJzdWIiOiJvY
2NtYXV0aHwxIiwiYXVkIjpbImh0dHBzOi8vYXBpLmNsb3VkLm5ldGFwcC5jb20iXSwiaHR0c
DovL2Nsb3VkLm5ldGFwcC5jb20vZnVsbF9uYW1lIjoiYWRtaW4iLCJodHRwOi8vY2xvdWQub
mV0YXBwLmNvbS9lbWFpbCI6ImFkbWluQG5ldGFwcC5jb20iLCJzY29wZSI6Im9wZW5pZCBwc
m9maWxlIiwiaWF0IjoxNjcyNzIyNzEzLCJleHAiOjE2NzI3NDQzMTMsImlzcyI6Imh0dHA6L
y9vY2NtYXV0aDo4NDIwLyJ9X_cQF8xttD0-S7sU2uph2cdu_kN-
fLWpdJJX98HODwPpVUitLcxV28_sQhuopjWobozPelNISf7KvMqcoXc5kLDyX-
yE0fH9gr4XgkdswjWcNvw2rRkFzjHpWrETgfqAMkZcAukV4DHuxogHWh6-
DggB1NgPZT8A_szHinud5W0HJ9c4AaT0zC-
sp81GaqMahPf0KcFVyjbBL4krOewgKHGFo_7ma_4mF39B1LCj7Vc2XvUd0wCaJvDMjwp19-
KbZqmmBX9vDnYp7SSxC1hHJRDStcFgJLdJHtowweNH2829KsjEGBTTcBdO8SvIDtctNH_GAx
wSgMT3zUfwaOimPw' \
> --header 'x-agent-id: vB_1xShPpBtUosjD7wfBlLIhqDgIPA0wclients' \
> -d '
> { "storage-server" : "sr630ip15.rtp.eng.netapp.com:10443", "access-
key": "2ZMYOAVAS5E70MCNH9", "secret-password":
"uk/6ikd4LjlXQOFnzSzP/T0zR4ZQlG0w1xgWsB" }'
```

Verify BlueXP backup and recovery settings

1. Select each ONTAP working environment and click **View Backups** next to the Backup and recovery service in the right-panel.

   You should be able to see all the backups that have been created for your volumes.

2. From the Restore Dashboard, under the Search & Restore section, click **Indexing Settings**.

   Make sure that the working environments which had Indexed Cataloging enabled previously remain enabled.

3. From the Search & Restore page, run a few catalog searches to confirm that the Indexed Catalog restore has been completed successfully.

Restart the BlueXP backup and recovery service

There may be situations where you'll need to restart the BlueXP backup and recovery service.

BlueXP backup and recovery functionality is built into the BlueXP Connector. You'll need to follow different initial steps to restart the service depending on whether you deployed the Connector in the cloud or whether you installed the Connector manually on a Linux system.

Steps

1. Connect to the Linux system that the Connector is running on.

Connector location	Procedure
Cloud deployment	Follow the instructions for connecting to the Connector Linux virtual machine depending on the cloud provider you're using.
Manual installation	Log in to the Linux system.

2. Enter the command to restart the service.

Connector location	Command
Cloud deployment	`docker restart cloudmanager_cbs`
Manual installation with internet access	`docker restart cloudmanager_cbs`
Manual installation without internet access	`docker restart ds_cloudmanager_cbs_1`

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